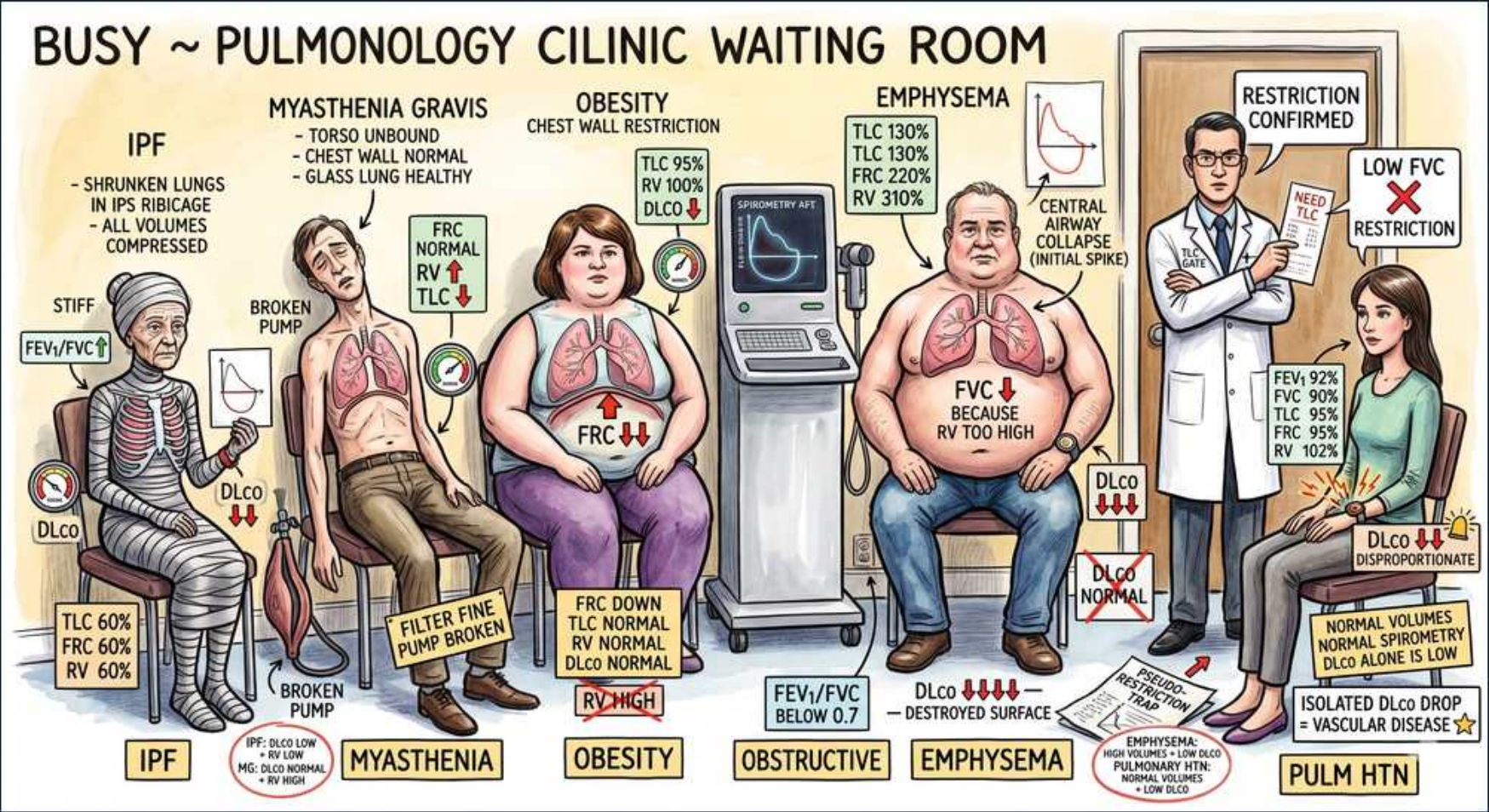




1. IPF / fibrosis patient

WHAT YOU SEE

Shrunken stiff figure with ribcage compressed, labeled IPF

WHAT IT ENCODES

Idiopathic pulmonary fibrosis = RESTRICTIVE: ↓TLC, ↓FRC, ↓RV, with a NORMAL or HIGH FEV1/FVC ratio (both numerator and denominator fall), and a LOW DLCO from membrane destruction. CT shows basal/subpleural reticulation, traction bronchiectasis, and honeycombing (UIP pattern); diagnosed >50–60 yrs, treated with antifibrotics (pirfenidone, nintedanib).

BOARD TRAP

Low DLCO + restriction = fibrosis/ILD, distinguishing it from chest-wall/neuromuscular restriction (NORMAL DLCO). DLCO often drops before lung volumes do, so it's the earliest marker.

2. IPF ↓DLCO tag

WHAT YOU SEE

DLCO ↓ icon under the IPF patient

WHAT IT ENCODES

DLCO is reduced in fibrosis because alveolar surface area is destroyed and the membrane is thickened — a diffusion limitation that also widens the A–a gradient and causes exertional desaturation.

BOARD TRAP

A NORMAL DLCO essentially argues against significant interstitial fibrosis — if restriction has a normal DLCO, look extrapulmonary.

3. Myasthenia gravis

WHAT YOU SEE

Figure with 'torso unbound, broken pump', labeled MYASTHENIA

WHAT IT ENCODES

Neuromuscular weakness (MG, ALS, Guillain-Barré, myopathy) = RESTRICTION with a NORMAL DLCO and normal FEV1/FVC; TLC and FVC fall, RV may rise (incomplete exhalation), and a supine FVC drop signals diaphragm weakness. Confirm MG with anti-AChR antibodies.

BOARD TRAP

Normal DLCO with restriction = neuromuscular/chest-wall cause, NOT parenchymal lung disease. Check maximal inspiratory/expiratory pressures (MIP/MEP) and supine FVC to localize.

4. Broken pump / filter fine

WHAT YOU SEE

Caption: 'filter fine, pump broken'

WHAT IT ENCODES

In neuromuscular disease the gas-exchange 'filter' (alveolar membrane) is intact while the muscular 'pump' fails. The result is hypoventilation: rising PaCO₂ with a NORMAL A–a gradient and preserved DLCO.

BOARD TRAP

Hypoxemia here is HYPOVENTILATION with a NORMAL A–a gradient, not a diffusion defect — and the treatment is ventilatory support (NIV), not just oxygen.

5. Obesity patient

WHAT YOU SEE

Heavy patient, labeled OBESITY, with arrows on FRC

WHAT IT ENCODES

Obesity: ↓FRC and ↓ERV first (chest-wall and abdominal mass), then mildly ↓TLC; FEV1/FVC is preserved and DLCO is NORMAL (or high). Can progress to obesity hypoventilation syndrome (daytime hypercapnia + OSA).

BOARD TRAP

Obesity gives a restrictive-LOOKING pattern but with a NORMAL/high DLCO — not fibrosis. The earliest change is a fall in ERV/FRC, not TLC.

6. FRC ↓ in obesity

WHAT YOU SEE

Red down-arrow on FRC under obese patient

WHAT IT ENCODES

Reduced FRC and ERV from chest-wall mass push tidal breathing toward closing volume, causing basal atelectasis and mild V/Q mismatch → mild hypoxemia that is WORSE supine.

BOARD TRAP

The EARLIEST obesity PFT change is a fall in ERV/FRC — TLC and the FEV1/FVC ratio stay relatively preserved until late.

7. Spirometry/plethysmograph

WHAT YOU SEE

Body box / spirometer machine center, FEV1/FVC below 0.7 label

WHAT IT ENCODES

Spirometry measures FEV1, FVC, and flows: FEV1/FVC <0.7 confirms OBSTRUCTION. To prove RESTRICTION you need lung volumes (TLC) by body plethysmography or gas dilution — spirometry alone cannot measure TLC, FRC, or RV.

BOARD TRAP

Spirometry diagnoses obstruction; a low FVC on spirometry only SUGGESTS restriction — you must confirm with a low TLC (plethysmography), or it may be pseudo-restriction/air trapping.

8. Emphysema patient

WHAT YOU SEE

Barrel-chest patient labeled EMPHYSEMA, TLC 130% RV 310%

WHAT IT ENCODES

Emphysema = OBSTRUCTION with hyperinflation: ↓FEV1/FVC, ↑TLC, ↑↑RV (here 310%), ↑FRC, and DLCO ↓↓↓ from destroyed alveolar surface area. Loss of elastic recoil drives air trapping; consider alpha-1 antitrypsin deficiency if early-onset or basilar/panacinar.

BOARD TRAP

Low DLCO separates emphysema from chronic bronchitis and asthma, where DLCO is NORMAL or high — the single best discriminator among obstructive diseases.

9. FVC ↓ from RV too high

WHAT YOU SEE

Caption: 'FVC↓ because RV too high'

WHAT IT ENCODES

In severe emphysema, air trapping raises RV so much that there's less volume left to exhale, so FVC FALLS — producing a low FVC that can masquerade as restriction (pseudo-restriction). But TLC is HIGH, not low.

BOARD TRAP

A low FVC in emphysema is air TRAPPING (high TLC), NOT coexisting true restriction (which requires a low TLC). Check TLC to tell them apart.

10. Central airway collapse spike

WHAT YOU SEE

Flow-volume tracing with initial spike, 'central airway collapse'

WHAT IT ENCODES

Emphysema's flow-volume loop shows a brief initial expiratory spike followed by rapid airway collapse and a markedly CONCAVE ('scooped') expiratory limb — loss of recoil collapses small airways early in exhalation.

BOARD TRAP

The scooped/concave expiratory limb signals diffuse SMALL-airway obstruction, NOT upper/central-airway obstruction (which gives flat plateaus).

11. Restriction confirmed (MD)

WHAT YOU SEE

Doctor with chart: FEV1 92%, TLC 60%, RV 60%, DLCO disproportionate

WHAT IT ENCODES

Restriction is CONFIRMED by a low TLC (here 60%) with a preserved/high FEV1/FVC ratio; once confirmed, DLCO localizes the cause (low = parenchymal ILD; normal = chest wall/neuromuscular/obesity).

BOARD TRAP

A low FVC with a normal ratio is restriction ONLY if TLC is also low — otherwise it's pseudo-restriction (air trapping/poor effort), so always confirm with lung volumes.

12. Pulmonary HTN patient

WHAT YOU SEE

Seated patient labeled PULM HTN, DLCO ↓ low

WHAT IT ENCODES

Pulmonary hypertension / pulmonary vascular disease: normal spirometry + normal lung volumes + an ISOLATED low DLCO (loss of the pulmonary capillary bed for CO uptake). Suspect with exertional dyspnea, loud P2, and a low DLCO out of proportion to other PFTs.

BOARD TRAP

Isolated low DLCO with normal volumes and normal spirometry points to VASCULAR disease (PAH/CTEPH), not parenchymal or airway disease.

13. Isolated DLCO drop sign

WHAT YOU SEE

Bottom-right sign: 'normal volumes, normal spirometry, DLCO alone low → vascular'

WHAT IT ENCODES

An isolated DLCO reduction with otherwise normal PFTs is the classic clue to pulmonary VASCULAR disease (PAH, chronic PE/CTEPH), and can also appear in EARLY ILD or with anemia (correct for Hgb first).

BOARD TRAP

Don't dismiss an isolated low DLCO as artifact — after Hgb correction it is a real, high-yield pointer to pulmonary vascular disease or early interstitial disease.